13524044 Narendra Dhowmo Wistaro M

K01

NO.

IF2120 Probabilitas dan Statistika

DATE

PR5 Hipotesis

10.17 A new curing process developed for a certain type of cement results in a mean compressive strength of 5000 kg/cm² with a std of 120 kg. To tes the hypotesis that $\mu = 5000$ against the alternative that $\mu < 5000$, a random sample of 50 piece of cement is tested. The critical region is defined to be $\bar{x} < 4970$

a) Probability of commiting a type I error when $H_0$ is true

b) evaluate $\beta$ for the alternative $\mu = 4970$ and $\mu = 4960$

a) $H_0 : \mu = 5000$

$H_1 : \mu < 5000$

tolak $H_0$ jika $\bar{x} < 4970$

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{120}{\sqrt{50}} \approx 16{,}97$

$Z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} = \frac{4970 - 5000}{16.97} = \frac{-30}{16{,}97} \approx -1{,}77$

$P(x < -1{,}77) \approx 0.0384$

$\therefore$ probabilitas dalam melakukan kesalahan tipe I adalah $0{,}0384 = 3{,}84\%$

b) $\mu = 4970$

$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{4970 - 4970}{16{,}97} = 0$

$P(Z \geq 0) = 0{,}5$

$\mu = 4960$

$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{4970 - 4960}{16{,}97} \approx 0{,}59$

$P(Z \geq 0{,}59) = 1 - P(Z < 0{,}59) \approx 0{,}7224$

$\beta = 1 - 0{,}7224 = 0{,}2776$

$\therefore \mu = 4970 \Rightarrow \beta = 0{,}5$ ; $\mu = 4960 \Rightarrow \beta = 0{,}2776$

10.21 An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative $\mu \neq 800$ hours, if a random sample of 30 bulbs has an average life of 788 hours. Use a P-value in your answer.

$H_0 : \mu = 800$

$H_1 : \mu \neq 800$

$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{788 - 800}{40/\sqrt{30}} = \frac{-12}{7{,}303} \approx -1.64$

$P(Z < |-1.64|) = P(Z < -1.64) + P(Z > 1.64)$

$= 2 \cdot P(Z < -1.64) = 2 \cdot 0.0505$

$= 0.1010$

Pilih $\alpha = 0{,}05$, maka $P > \alpha$, maka gagal tolak $H_0$. Tidak ada bukti cukup kuat untuk menyatakan rata² berbeda dari 800 jam

LOOSE LEAF L1200 7mm×31lines maruman

Dipindai dengan CamScanner

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10.41 A study was conducted by the Department of Zoology at Virginia Tech to determine if there's a significant difference in the density of organisms at two different stations. The following data give the density measurements in a number of organisms per $m^2$.

| Station 1 | | Station 2 | |
|---|---|---|---|
| 5030 | 4980 | 2800 | 2810 |
| 12 200 | 11.910 | 4670 | 4770 |
| 10.770 | 2170 | 6870 | 3720 |
| 11.400 | 26850 | 7720 | 1720 |
| 860 | 17660 | 7010 | 4190 |
| 2200 | 22800 | 7330 | 2190 |
| 4250 | 1130 | | |
| 15 040 | 1690 | | |

Can we conclude at the 0.05 significance, that the average densities at the two stations are equal? Assume that the observations come from normal population with a different variance.

Opsions : $\sigma_1^2 \neq \sigma_2^2$ (Welch's T-test)

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

Station 1 :

$\Sigma x_1 = 152.360$

$n = 16$

$\bar{x}_1 = 9.897,5$

$S_1^2 = \frac{\Sigma(x_1 - \bar{x})^2}{n-1} = 61\,862.633,33$

$S_1 = 7.861,46$

Station 2

$n_2 = 12$

$\Sigma x_2 = 49.950$

$\bar{x}_2 = \frac{49950}{12} \approx 4120,83$

$S_2^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1} = 5\,625.633,42$

$S_2 = 2371,85$

$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{9897,5 - 4120,83}{\sqrt{\frac{61\,862.633,33}{16} + \frac{5625.633,42}{12}}}$

$= \frac{5.776,67}{\sqrt{3.867.664,58 + 468.806,29}} = \frac{5.776,67}{2081,21} \approx 2,776$

$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}} = \frac{3.867.664,58 + 468.806,29}{\frac{3.867.664,58^2}{15} + \frac{468.806,29^2}{11}}$

$= \frac{1,867 \times 10^{13}}{1,014 \times 10^{12}} \approx 18,5$

gunakan $v = 19$

lo $t'_{crit} = 2,101$

karena $t_{hitung} > t_{crit}$, maka tolak $H_0$.

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NO.

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PR 5 Latihan Hipotesis

10.61 In a survey of an epidemic flu, the parents of 2000 babies were surveyed by researchers of a well-known pharmaceutical company to determine if the company's new drug was effective after two days. Among babies who had flu but were not given the medicine, 56 recovered within two days. Is there any significant indication that supports the company's claim of the effectiveness of the medicine?

$H_0 : p_1 = p_2$

$H_1 : p_1 > p_2$

$n_1 = 120 \quad x_1 = 29 \quad p_1 = \frac{29}{120} \approx 0,2417$

$n_2 = 280 \quad x_2 = 56 \quad p_2 = \frac{56}{280} = 0,2$

$\hat{p} = \frac{29 + 56}{120 + 280} = 0,2125$

$\hat{q} = 1 - \hat{p} = 0,7875$

$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0,2417 - 0,2}{\sqrt{0,2125 \cdot 0,7875\left(\frac{1}{120} + \frac{1}{280}\right)}}$

$\approx 0,935$

$P(z < 0,935) \approx 0,8251$

$P(z > 0,935) = 1 - 0,8251 = 0,1749$

Karena $P > \alpha \Rightarrow 0,1749 > 0,05$, maka gagal tolak $H_0$.

10.77 An experiment was conducted to compare the alcohol content of soy sauce on 2 different production lines.

PL 1 :

0.48 0.39 0.41 0.52 0.40 0.40 0.52 0.5

PL 2

0.38 0.37 0.39 0.41 0.38 0.39 0.40 0.3

Assume both populations are normal. It is suspected that PL 1 isn't producing consistently as PL 2 in terms of alcohol content. Test the hypothesis that $\sigma_1 = \sigma_2$ against alternative $\sigma_1 \neq \sigma_2$ use a P-value

$H_0 : \sigma_1 = \sigma_2$

$H_1 : \sigma_1 \neq \sigma_2$

| PL 1 | PL 2 |
|---|---|
| $n_1 = 8$ | $n = 8$ |
| $\bar{x}_1 = 0,466$ | $\bar{x}_2 = 0,389$ |
| $S_1^2 \approx 0,003055$ | $S_2^2 = 0,000155$ |

$F = \frac{S_1^2}{S_2^2} = \frac{0.003055}{0,000155} \approx 19,67$

$v_1 = n - 1 = 7$

$v_2 = n - 1 = 7$

karena dua sisi, 2 pihak, maka

$P = 2P(F_{(7,7)} > 19,67) = 2 \times 0,00042$

$= 0,00084$

$\alpha = 0,05$, karena $P < 0,05$

maka tolak $H_0$.

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10.93) To determine current attitudes about prayer in public schools, a survey was conducted in four Virginia counties. The following table gives the attitudes of 260 parents from Craig County, 150 parents from Giles County, 100 parents from Franklin County and 100 parents from Montgomery County

| Attitude | Craig | Giles | Franklin | Mont. |
|---|---|---|---|---|
| Favor | 65 | 66 | 40 | 34 |
| Oppose | 42 | 30 | 33 | 42 |
| No opinion | 93 | 54 | 27 | 24 |

Test for homogeneity of attitudes among the four counties prayer in the public schools. Use a P value in your conclusion

$H_0$ : homogen

$H_1$ : tidak homogen

| $E_{ij}$ | Craig | Giles | Franklin | Mont. | Total Baris |
|---|---|---|---|---|---|
| Favor | 74,55 | 55,91 | 37,27 | 37,27 | 205 |
| Opposed | 53,45 | 40,09 | 26,73 | 26,73 | 147 |
| No opinion | 72 | 54 | 36 | 36 | 198 |
| Total kolom | 260 | 150 | 100 | 100 | 550 |

$$E_{ij} = \frac{\text{total baris} \times \text{total kolom}}{\text{total tabel}}$$

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \approx 31{,}09$$

$$v = (r-1)(c-1) = (3-1)(4-1) = 6$$

untuk $v = 6$ dan $\alpha = 0{,}05 \Rightarrow \chi_0^2 = 12{,}592$

karena $\chi^2 > \chi_0^2 \Rightarrow 31{,}09 > 12{,}592$

10.109. The following data show of defects in 100.000 lines of code in a particular type of software program developed in US and Japan. Is there enough evidence to claim that there is a significant difference between programs developed in two countries? Test at $\alpha = 0{,}05$. Should variance be pooled?

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| U.S. | 48 | 39 | 42 | 52 | 40 | 48 | 52 | 52 |
| | 54 | 48 | 52 | 55 | 43 | 46 | 48 | 52 |
| Japan | 50 | 48 | 42 | 40 | 43 | 48 | 50 | 46 |
| | 38 | 38 | 32 | 40 | 40 | 48 | 48 | 45 |

b) $H_0 : \sigma_1^2 = \sigma_2^2$

$H_1 : \sigma_1^2 \neq \sigma_2^2$

$n_1 = 16 \quad \bar{x}_1 = 48{,}1875 \quad s_1^2 = 24{,}029$

$n_2 = 16 \quad \bar{x}_2 = 43{,}75 \quad s_2^2 = 22{,}067$

$v_1 = 15 \quad v_2 = 15$

$$F = \frac{s_1^2}{s_2^2} = \frac{24{,}029}{22{,}067} = 1{,}089$$

untuk $\alpha = 0{,}05$, nilai $F_{\alpha, 15, 15} = 2{,}86$

karena $F_{hitung} < F_{kritis} \Rightarrow 1{,}089 < 2{,}86$

maka gagal tolak $H_0$. oleh karena itu varians harus digabung (pooled)

a) $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$$S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(15)(24{,}029) + (15)(22{,}067)}{16 + 16 - 2}$$

$= 23{,}048$

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$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{48{,}1875 - 43{,}75}{4{,}80\sqrt{\frac{1}{16} + \frac{1}{16}}} = 2{,}615$$

dengan $df = n_1 + n_2 - 2 = 30$ dan $\alpha = 0{,}05 \Rightarrow \alpha/2 = 0{,}025$

maka nilai kritisnya adalah $\pm 2{,}042$

karena $t_{hitung} > t_{kritis}$, maka tolak $H_0$.

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